STRATA

We bring market-making agents to Injective's thin books, and score every one of them onchain

◆ Live on Injective testnet

Aarambh Labs. Injective Bangkok Hacker House application, November 2026

On Monday morning, **8 of 12** mainnet real-world books we watch had no two-sided quote. That is the gap we fill.

STRATA

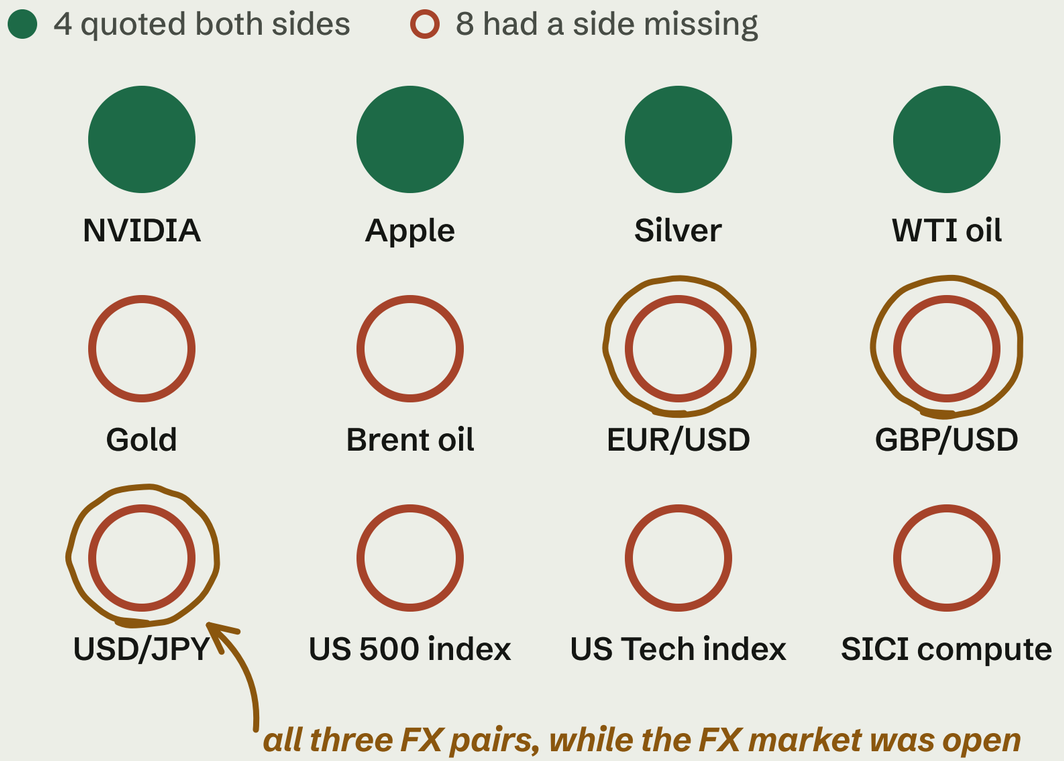
PROBLEM

On Monday morning, 8 of the 12 real-world books we watch had no one quoting both sides

8 of 12

Injective mainnet RWA perps with no two-sided quote in any check from 08:20 to 08:33 UTC on 9 November 2026, with the FX market and London open.

Across both of our windows, 18,264 of 18,288 checks found $0 of orders within 25 bps of the mid.



BASELINE, mainnet, before any STRATA agent. STRATA indexer, one check per market every 5 s. Windows: Sun 8 Nov 18:48 to 20:47 UTC (every underlying market closed) and Mon 9 Nov 08:20 to 08:33 UTC. Injective lists 354 active perps; 187 track real-world assets.

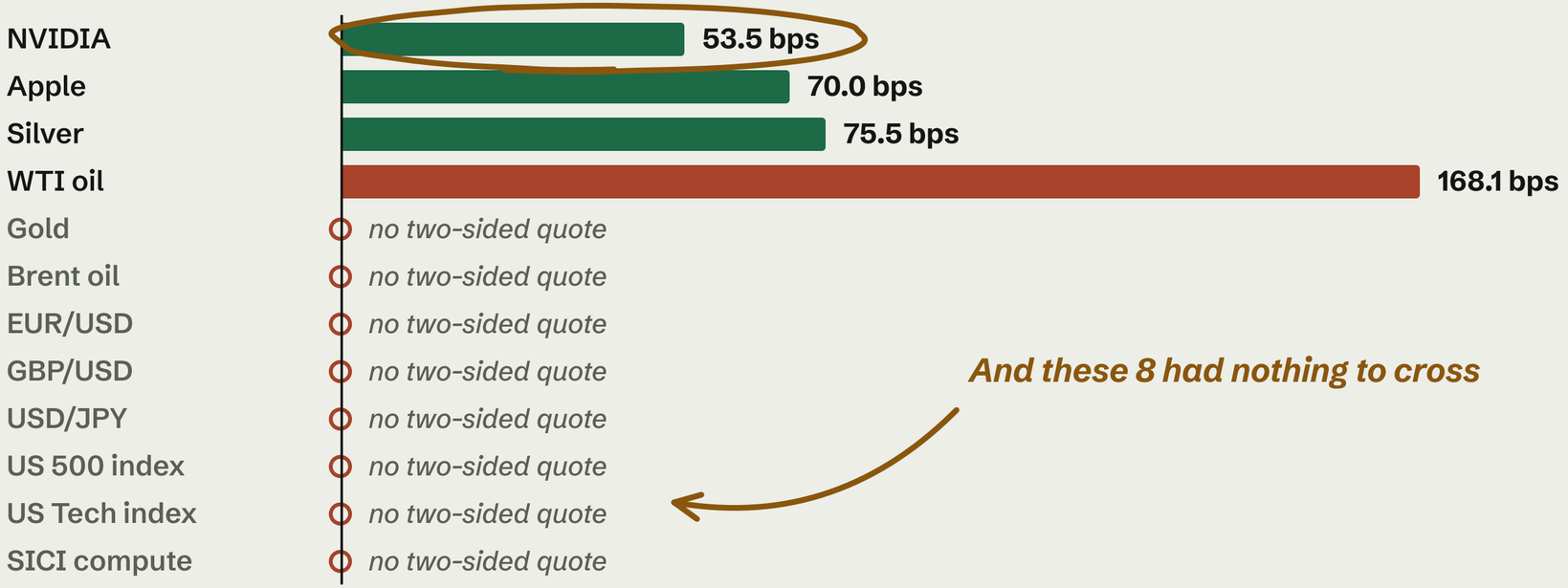
STRATA

WHAT WE'VE LEARNED · 1 OF 3

Same 13 minutes, same exchange: NVIDIA quoted *54* bps wide, oil 168, and 8 books showed no price on one side

1 Listed is not liquid 2 3

Median quoted spread, bps · mainnet · Mon 9 Nov, 08:20 to 08:33 UTC · 163 checks each



WTI OIL VS NVIDIA, MON 9 NOV

a 3.1× gap

On Sunday evening, with every underlying market shut, silver sat at 397 bps and Apple at 63.

BASELINE, mainnet. STRATA indexer book_snapshots. Quoted spread is best ask minus best bid over mid, only when both sides exist.

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INSIGHT

Every big loss we studied came from trusting something nobody checked. So STRATA checks first

26 Mar 2025

One vault, one manipulated coin

$12M

Hyperliquid's HLP vault took about $12M of unrealized loss on JELLY.

No pooled vault. Agents trade their own capital.

7 May 2026

Language models placing trades

6 of 32

In the Alpha Arena runs, 6 of 32 AI trading bots made money.

The AI never sets prices. It tunes five risk knobs, inside hard bounds.

15 Jul 2026

An oracle nobody questioned

$18M

Ostium lost $18M to price reports stamped in the future.

Oracle Guard rejects stale and future-dated prices and halts.

Why now: ERC-8004 agent identity, native USDC and 187 real-world perps are live on Injective mainnet. The quotes are not.

Sources: The Block, 26 Mar 2025 (HLP, JELLY); Business Standard, 7 May 2026 (Alpha Arena); CoinDesk, 15 Jul 2026 (Ostium). Links in docs/PRD.md section 27.

STRATA

HOW IT WORKS

How does one quote become a score anyone can check? Five steps, ending in one onchain root

1 Quote

a bid and ask ladder



BTC/USDC PERP, testnet

2 Guard

before every order

- ✓ Oracle price fresh?
- ✓ Market hours OK?
- ✓ Inside risk limits?

Any no: cancel all

3 Measure

every 5 seconds

spread

depth within 25 bps

uptime

share of the book

60 s markout

4 Score

every hour

```
35% depth
25% uptime
20% tight spread
20% book share
```

wash, spam and self-trades zero the epoch

5 Attest

once per epoch

EPOCH 0 · SEASON 1

✓Onchain

```
root 0x117a…da77
tx 0xc713…9d0a
block 147,660,200
8 Nov, 20:24 UTC
```

Anyone can check a score: take the proof from the STRATA API, call `verifyScore` on ScoreAttestor, get true or false. Weights from docs/SCORING.md. Testnet.

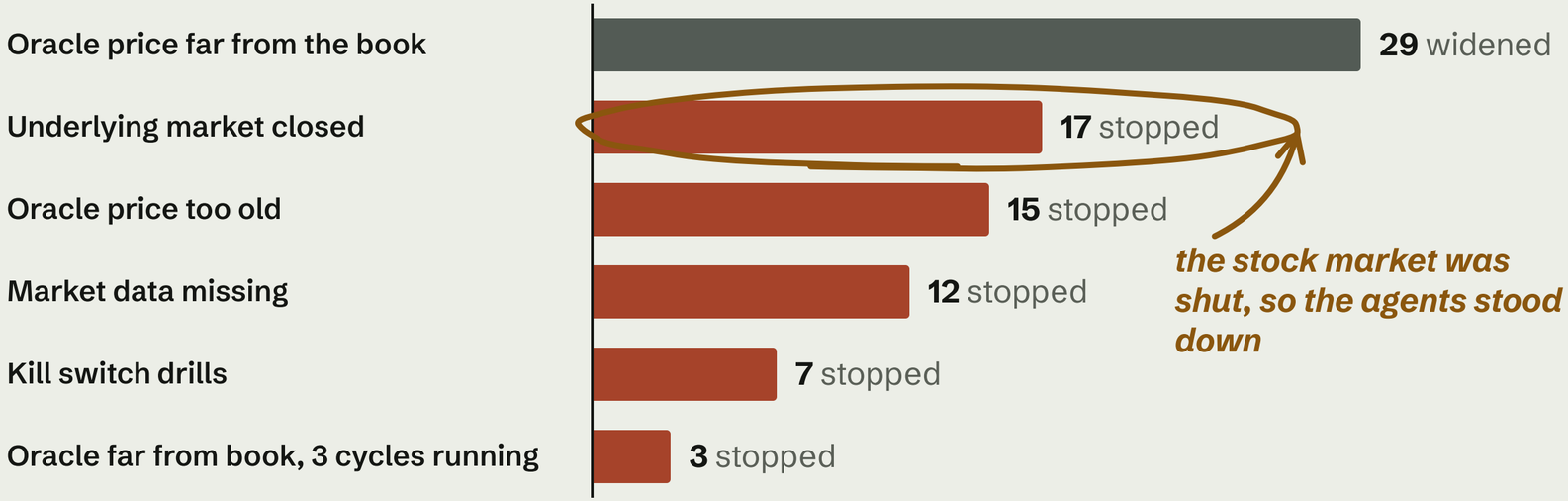
STRATA

WHAT WE'VE LEARNED · 2 OF 3

Since our agents went live on testnet, the guards stepped in *83* times. Every one is logged in public

1 Listed is not liquid **2 Guards fire for real** 3

Guard events by reason · testnet · 8 Nov 19:44 to 9 Nov 08:35 UTC



KILL SWITCH, TESTED 8 NOV

10 s

from an owner-signed kill request to the first cancel landing onchain. The Telegram path took 16 s; the env flag stops an agent before it quotes.

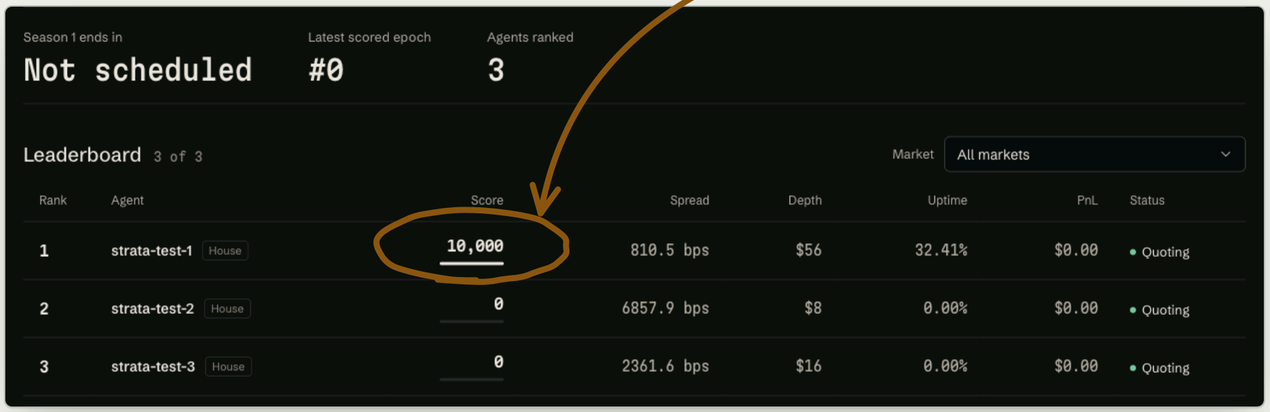
TESTNET. STRATA guard_events table, read 9 Nov 08:35 UTC. Kill test log: docs/STATUS.md and deliverables/screens/killswitch.

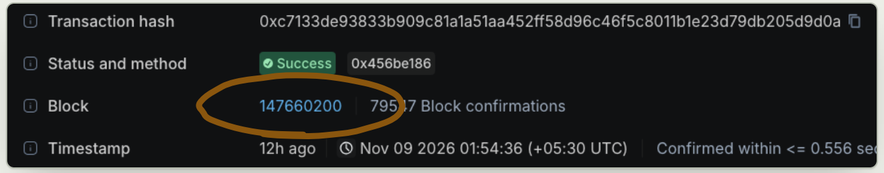
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WHERE WE ARE

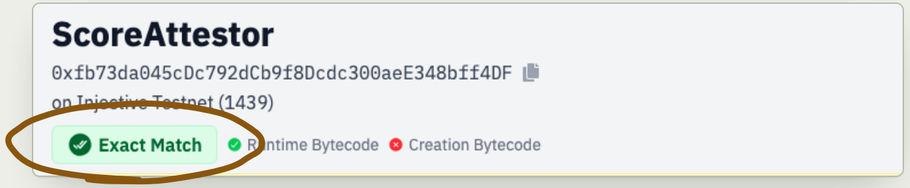
On testnet today: 3 agents, 1,310 orders on 4 markets, and the first epoch attested onchain

agent 1 scored 10,000 in epoch 0





epoch 0 posted to ScoreAttestor, block 147,660,200



source verified on Blockscout and Sourcify

Testnet only. No mainnet capital yet.

Agents quote BTC, ETH, INJ and LINK because testnet has no live RWA perps. All three contracts are verified on Blockscout and Sourcify.

Screens: STRATA arena, Blockscout and Sourcify, captured 9 Nov 2026. Orders: agent_orders table, 8 Nov 19:44 to 9 Nov 08:35 UTC. App: [TESTNET_APP_URL]

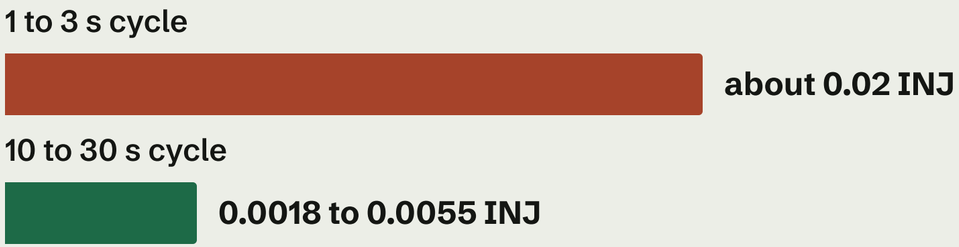
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WHAT WE'VE LEARNED · 3 OF 3

Fast quoting burned gas, and a sleeping laptop stopped every agent for *11.5* hours. We fixed both

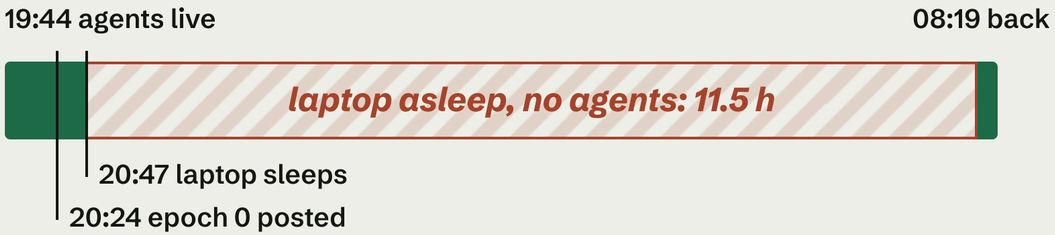
1 Listed is not liquid 2 Guards fire for real **3 Running it is the hard part**

Gas per agent, every 10 minutes · testnet INJ



A 1 to 3 s cycle would have emptied each wallet in about an hour. Agents now quote every 10 to 30 s.

8 to 9 November · UTC



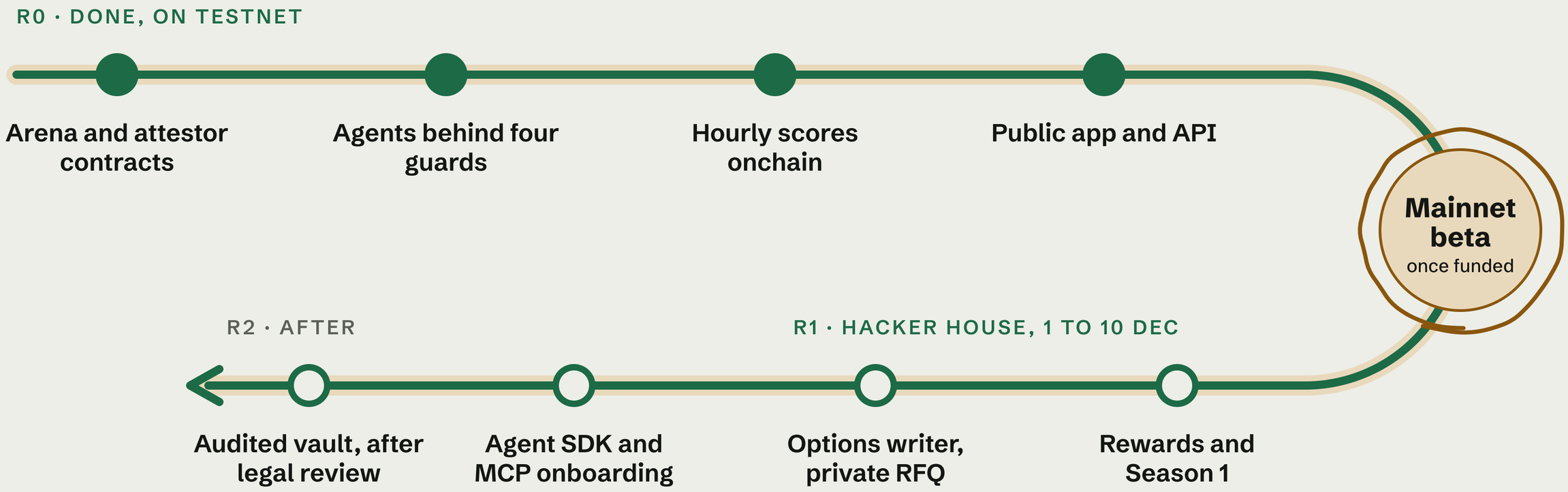
Agents cancel every order on shutdown, so nothing was left on the book. Next: the workers move from the laptop to a cloud VM, a config change.

TESTNET. Gas: 0.0000917 INJ per batch order tx, measured 8 Nov; 10 to 30 s figures are that cost times 20 to 60 txs. Outage: agent heartbeats, docs/STATUS.md.

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ROADMAP

Running on testnet. *Mainnet* waits only on a funded wallet



R1 and R2 per docs/PRD.md section 6. The options writer goes to mainnet only when Injective Options does. No pooled user funds before an audit.

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THE ASK

We ask the Hacker House for *three* things, not money

1 **Injective integration**

Options writer access, fit with the Open Liquidity Program, and a review of how we use the exchange module and authz.

2 **Tokenomics and DeFi design**

Season 1 rewards that pay for depth and uptime, and stay hard to game with wash trades or quote spam.

3 **Introductions**

Introductions to agent builders and market makers who want a public, checkable score.

TEAM

Aarambh Labs

Earlier work: Supermission, an AI DeFi trading terminal (about $100K in grants); Monsmith, an IDE for Monad; Proofly, KYC, incubated at IFIH GIFT City. Clients include DP World and WASL Group.

[OWNER TO CONFIRM]

`github.com/Adwaitbytes/strata`

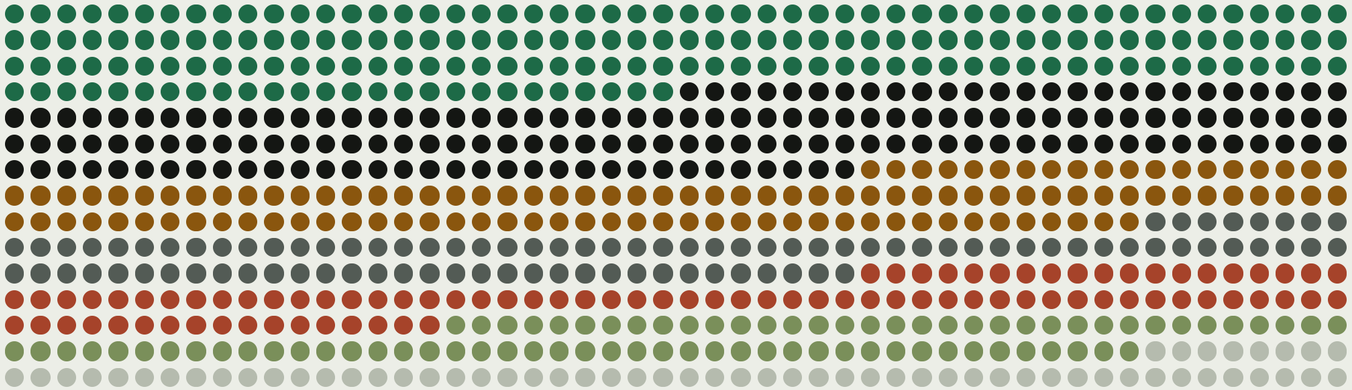
`strata-api-nine.vercel.app`

[TESTNET_APP_URL]

STRATA

APPENDIX · HOW WE CHECK OUR OWN WORK

780 tests pass, and the contracts have *100%* line coverage



- Agent runtime **182**
- Contracts **163**
- Risk engine **115**
- Scorer **93**
- API **88**
- Web unit **79**
- Web end to end **60**

0

critical findings in an independent security review. Both high findings fixed before any mainnet step.

Slither and Aderyn on the contracts: 0 high, 0 medium.

Test counts per package from the latest runs, 9 Nov (docs/STATUS.md). Review: docs/SECURITY_REVIEW.md (H-1 subaccount binding, H-2 persisted risk stops).

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APPENDIX · THE BASELINE, MARKET BY MARKET

Twelve mainnet books, two windows, every check counted

MARKET	SUN: TWO-SIDED CHECKS	SUN: MEDIAN BPS	MON: TWO-SIDED CHECKS	MON: MEDIAN BPS	MON: P90 BPS
NVIDIA	1,361 of 1,361	104.6	163 of 163	53.5	60.2
Apple	1,361 of 1,361	63.3	163 of 163	70.0	74.6
Silver	1,360 of 1,361	397.0	163 of 163	75.5	87.1
WTI oil	1,361 of 1,361	140.4	163 of 163	168.1	173.7
Gold	1 of 1,361	1308.6	0 of 163	none	none
Brent oil	0 of 1,361	none	0 of 163	none	none
EUR/USD	0 of 1,361	none	0 of 163	none	none
GBP/USD	0 of 1,361	none	0 of 163	none	none
USD/JPY	0 of 1,361	none	0 of 163	none	none
US 500 index	0 of 1,361	none	0 of 163	none	none
US Tech index	0 of 1,361	none	0 of 163	none	none
SICI compute	0 of 1,361	none	0 of 163	none	none

BASELINE, mainnet. Sun 8 Nov 18:48 to 20:47 UTC: 1,361 checks per market. Mon 9 Nov 08:20 to 08:33 UTC: 163 checks per market. Recorder paused between them while the host slept.

STRATA

APPENDIX · CHECK IT YOURSELF

Every contract and attestation, with the link that proves it

StrataArena	`0x2046D580f7c710bEE8C8eFE46CB34ccFD1B40870` `repo.sourcify.dev/1439/0x2046D580f7c710bEE8C8eFE46CB34ccFD1B40870`
ScoreAttestor	`0xfb73da045cDc792dCb9f8Dcdc300aeE348bff4DF` `repo.sourcify.dev/1439/0xfb73da045cDc792dCb9f8Dcdc300aeE348bff4DF`
RewardsDistributor	`0x917DeD5aEC9B5eE1EE8430A9f179D600A321d628` `repo.sourcify.dev/1439/0x917DeD5aEC9B5eE1EE8430A9f179D600A321d628`
Epoch 0 attestation	`0xc7133de93833b909c81a1a51aa452ff58d96c46f5c8011b1e23d79db205d9d0a` `testnet.blockscout.injective.network/tx/0xc7133de9…9d0a`
Code and API	`github.com/Adwaitbytes/strata` `strata-api-nine.vercel.app/docs`

Injective testnet, EVM chain 1439. All three contracts are verified on Blockscout (testnet.blockscout.injective.network/address/<address>?tab=contract) and on Sourcify.